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/20	/20	/20	/20	/20	/20	/100

Name _____

April 26, 2008 Saturday

Department _____

SID _____

University Physics
Midterm Examination
Kuang Yaming Honors School, Nanjing University

Physical Constants

Gravitational constant

$$G = 6.67 \times 10^{-11} \text{ m}^3 / \text{s}^2 \cdot \text{kg}$$

Standard gravity acceleration

$$g_n = 9.806 \text{ m/s}^2$$

Select five out of following six problems.

(1) (20pts) A block of mass m moves on a horizontal surface. The friction force acted on the block is $f = \alpha + \beta v^2$ where α and β are constants. The initial velocity of the block is v_0 .

- Write the equation of motion of the block.
- After what time interval does the block come to rest?
- What distance has then been covered by the block?

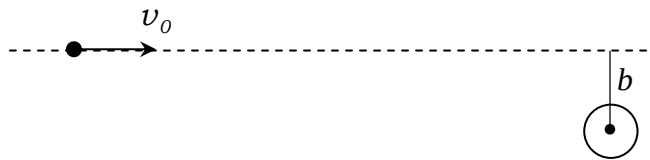
(2) (20pts) A particle of mass m moves in one dimension has a potential energy

$$U(x) = U_0 \left[2 \left(\frac{x}{a} \right)^2 - \left(\frac{x}{a} \right)^4 \right]$$

where U_0 and a are positive constants.

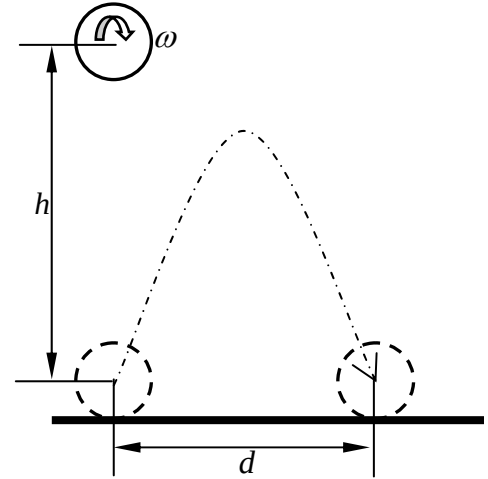
- Find the force $F(x)$, which acts on the particle.
- Find the positions of stable and unstable equilibrium.
- What is the minimum speed the particle must have at the origin to escape to infinity?

(3) (20pts) A space probe heads towards a very distant planet of radius R and very large mass M with initial velocity v_0 and impact parameter b .



- (a) Find the maximum value of the impact parameter b for which the probe will hit the planet.
- (b) What is the velocity of the probe when it just grazes the planet?

(4) (20pts) If a ball of mass m is dropped from rest at a height h from a horizontal ground, it bounces back to a height of kh where $0 < k < 1$ after it hits the ground. If the ball is dropped again from the same height with an initial angular velocity ω , it bounces and drops to the ground a distance d apart from the point it first hits the ground, as shown in the accompanying figure. Assume h and d are much larger than the radius of the ball.



- What is the impulse I on the ball from the collision with the ground when it is dropped from rest?
- What is the kinetic frictional coefficient μ between the surface of the ball and the ground?

(5) (20pts) A uniform solid cylinder of mass m and radius R is at rest on a frictionless horizontal surface. The cylinder is struck horizontally at a point on its circumference that is a height h above the surface. Immediately after this instantaneous hitting, the cylinder begins to roll without slipping at a velocity v_0 . And the cylinder then encounters an inclined plane that makes an angle α with the horizontal. The surface of the inclined plane is also frictionless.



- Find the rotational inertia of cylinder about its center of mass. Find the height h of the hitting point.
- Does the cylinder continue to roll without slipping upon the inclined plane? Why?
- What is the maximum height $H_{\max}$ that the cylinder can reach on the inclined plane?

(6) (20pts) As shown in the figure, a thin circular hoop is suspended at its top in a way so that it can oscillate freely in the vertical plane of the hoop. The mass of the hoop is M and the radius of it is R .

- (a) Write down the equation of rotational motion of the hoop.
- (b) Find the period of small oscillations about its equilibrium position.
- (c) If the hoop oscillates in the direction perpendicular to its plane, what is the period of small oscillations?

